

Risk Assessment

Criteria	Level	Category
1 = Low	1 = Very Low	Strategic
2 = Medium	2 = Low	Operational
3 = High	3, 4 = Medium	Financial
	6 = High	External
	9 = Critical	

Risk ID	Description	Risk Category	Probability	Impact	Level	Strategy	Treatment
R-001	AI credits are too expensive to build and test the AI scenario	Financial	3	3	9	Avoid	Use a self-hosted LLM, check if the BFH can provide AI services or if there are free plans for education
R-006	Scope is too large for bachelor thesis	Strategic	2	3	6	Mitigate	We have to plan and evaluate the goals carefully. The use of "User Story Maps" also helps to mitigate this risk.
R-007	Scenario or drill too complex for automation	Strategic	2	3	6	Mitigate	Create "User Story Maps" for epics, so that we have a good understanding of the scenarios and their drills.
R-002	Not enough time for documentation	Operational	2	3	6	Mitigate	Add a dedicated documentation issue with an according weight to each epic.
R-003	Not enough time for testing (drills, scenarios and environments)	Operational	3	2	6	Mitigate	Add a dedicated testing issue with an according weight to each epic.
R-004	Simulating or building network environments needs more time than planned	Operational	3	2	6	Mitigate	Add dedicated issues with according weight to each scenario epic.
R-005	User does not understand how to operate NSAK	Operational	3	2	6	Mitigate	Provide a Web GUI and/or user guides
R-012	Misaligned communication leads to wrong expectations	Strategic	2	2	4	Mitigate	Multiple channels and Bi-Weekly Meetings should keep the communication aligned.
R-010	Scenarios and drills need more time than planned	Operational	2	2	4	Mitigate	Use "User Story Maps" to get a good idea of the required steps to build the scenario, it's drills and estimate generously.
R-011	Not enough time for media artifacts (poster, presentations, film, book, etc.)	Operational	2	2	4	Mitigate	Include the creation of the media artifacts as milestones/epics in the project planning.
R-009	Complex network environments need more hardware than available	Financial	2	2	4	Avoid	We can simulate network environments instead of real hardware.
R-008	The AI bubble bursts and we need to switch AI provider	External	2	2	4	Avoid	Use vendor agnostic tools and protocols, so that we can switch provider easily.
R-014	Regressions: tested scenarios stop working because a drill was changed	Operational	3	1	3	Mitigate	Add automated test cases for detecting regressions early.
R-015	The NSAK framework/device itself is vulnerable	Operational	3	1	3	Transfer	Clearly state that the NSAK framework/device is still a POC and not ready for production use.
R-013	One of the students gets injured or sick	External	1	3	3	Accept	We accept this risk.
R-019	An existing framework already does everything we have planned	Strategic	2	1	2	Accept	We accept the risk, but will compare well known frameworks with NSAK and highlight our USP (HW and AI).
R-017	Drills not compatible between scenarios	Operational	2	1	2	Mitigate	The implementation of the configuration management should provide a way that small deviations can be branched.
R-018	The AI might go rogue in the AI scenario	Operational	2	1	2	Mitigate	Besides the fact that we have physical access to the device we add a killswitch to the AI scenario.
R-016	The NSAK hardware devices stop working	Financial	1	2	2	Accept	The fact that we have two devices already reduces the impact, further we could also use other HW.
R-021	Lack of resources for attack scenarios	Operational	1	1	1	Mitigate	Get familiar and use the MITRE ATT&CK framework as reference and source.
R-022	Lack of resources / consensus for defense scenarios	Operational	1	1	1	Mitigate	Get familiar and use well known sources such as NIST as source.
R-020	NSAK gets abused by hackers	External	1	1	1	Transfer	Clearly state that the NSAK framework/device is built only for educational and white hate purposes.